Bivariate statistics

Lecture and worked examples

Maths Explorer

Reference course

A line of best fit $y = ax + b$ gives estimates, not exact equalities. The coefficient a describes the estimated average change in y when x increases by one unit. Interpolation stays within the range of observed values; extrapolation goes outside that range and must be presented with caution.

Reference course

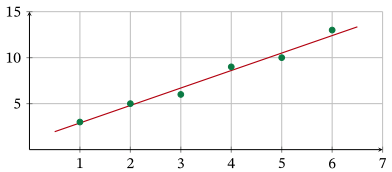
To use $y = ax + b$: 1) replace x by the requested value to estimate y; 2) solve $ax + b = y_0$ to estimate x; 3) round according to the context; 4) state whether it is interpolation or extrapolation; 5) remind that the result depends on the model.

Scatter plot

- Construct a scatter plot.
- Choose the scales.
- Identify the variables.

Scatter plot

The table (x_i, y_i) gives the advertising spending and the sales over six months:
$(1, 12), (2, 15), (3, 17), (4, 22), (5, 24), (6, 28)$. Choose scales, draw the scatter plot and plot the mean point.



Scatter plot

A scatter plot represents the pairs (x_i, y_i). The axes, units and scales must be indicated; each pair gives a single point.

Scatter plot

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Scatter plot

One-hour plan: 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.

Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.

Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Reading and interpreting a scatter plot

- Describe a trend.
- Spot spread and outliers.
- Avoid unwarranted causality.

Reading and interpreting a scatter plot

Compare three scatter plots: a tight increasing trend, a scattered decreasing trend, no trend. For each, write two sentences without using the word “cause”.

Reading and interpreting a scatter plot

Describing a scatter plot means commenting on its direction, shape, spread and outliers, without claiming a causality that has not been established.

Reading and interpreting a scatter plot

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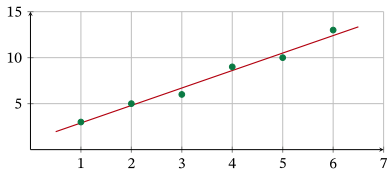
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Linear regression

- Obtain a linear regression.
- Interpret a and b.
- Check with the mean point.

Linear regression

With the data of session 31, obtain the linear regression with a calculator. Interpret the slope and check that the mean point lies approximately on the line.



Linear regression

A line $y = ax + b$ fits a scatter plot. a represents the estimated average change in y for one extra unit of x.

Linear regression

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Linear regression

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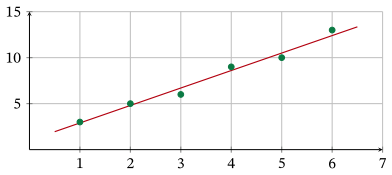
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Interpolation

- Interpolate.
- Solve using the model.
- Express an estimate.

Interpolation

Over the observed period $x \in [1, 12]$, the model is $y = 2.8x + 14$. Estimate y for $x = 7.5$, then estimate x when $y = 35$. Classify each estimate.



Interpolation

An interpolation estimates a value inside the observed range. The result is an estimate that depends on the model.

Interpolation

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Interpolation

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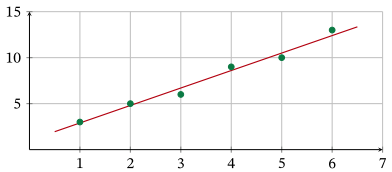
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Extrapolation

- Extrapolate with caution.
- Assess plausibility.
- Present a limitation of the model.

Extrapolation

The same model $y = 2.8x + 14$, fitted on $[1, 12]$, is used for $x = 40$. Calculate the estimate, then give three reasons why it may be unreliable.



Extrapolation

An extrapolation estimates outside the observed range. The further away we go, the more caution is needed.

Extrapolation

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Extrapolation

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Change of variable

- Carry out a change of variable.
- Fit the new data.
- Return to the original variable.

Change of variable

The data appear to be exponential. We set $z = \log(y)$ and the fit gives $z = 0.08x + 1.2$. Express y as a function of x, then calculate y for $x = 5$.

Change of variable

A change of variable can make a scatter plot linear. After fitting in the new variable, we must return to the original variable.

Change of variable

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Change of variable

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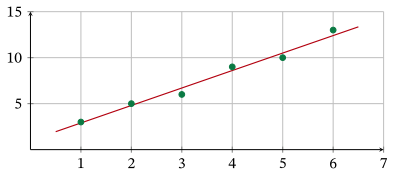
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Choosing the model and fitting

- Compare models.
- Use residuals.
- Define a range of use.

Choosing the model and fitting

Two models give total absolute residuals of 18.4 and 11.7 over the observed range. The second, however, shows unrealistic growth beyond $x = 20$. Choose a model to interpolate at $x = 12$, then discuss extrapolation at $x = 50$.



Choosing the model and fitting

We choose a model according to the shape of the scatter plot, the size of the residuals and the range of use. A better visual fit does not allow unlimited extrapolation.

Choosing the model and fitting

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Review and assessment

- Choose and use a regression.
- Write a statistical conclusion.
- Analyse an error.

Review and assessment

Starting from a scatter plot supplied by the teacher, produce a complete answer: choice of model, equation, estimate, type of estimate, unit and caveat. Swap papers and use a six-criterion grid.

Review and assessment

A complete statistical answer contains the model, the calculation, the rounding, the unit, the type of estimate and a caveat on validity.

Review and assessment

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